

EBOOK MOTION SYSTEM

Robot Humanoid Gandamana – Universitas Negeri Surabaya

KRSBI-H (Kontes Robot Sepak Bola Indonesia - Humanoid)

Platform: ROBOTIS OP3 | ROS Package: kri2027 | Tahun: 2025-2026

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1. PENDAHULUAN

1.1 Tentang Dokumen Ini

Ebook ini adalah referensi lengkap untuk **Motion System** robot humanoid Gandamana yang berlaga di KRSBI-H. Seluruh kode sumber berada di folder `motion/src/` sebagai bagian dari ROS package `kri2027`.

1.2 Hardware Platform

Komponen	Spesifikasi
Robot	ROBOTIS OP3
Main Controller	Intel NUC / Onboard Computer (ROS)
Sub-Controller	OpenCR (STM32F4)
Aktuator	Dynamixel MX-28/XL-320 (20 servo)
DOF Kaki	6 per kaki (hip yaw/roll/pitch, knee, ankle pitch/roll)

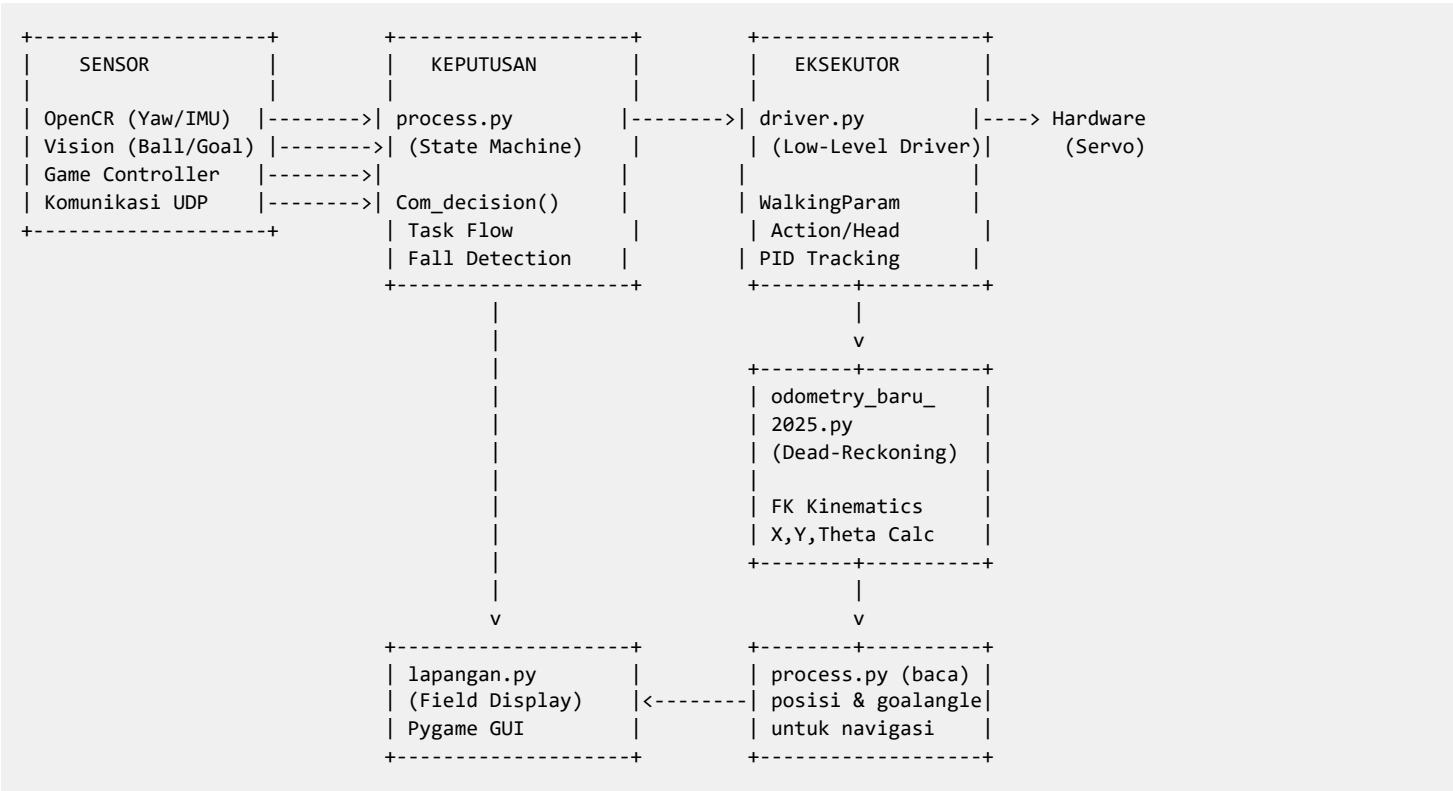
Komponen	Spesifikasi
DOF Kepala	2 (pan + tilt)
Kamera	320x240 @ 60 FPS (V4L2)
IMU	BNO055 (di dalam OpenCR)
Yaw Sensor	Magnetometer (OpenCR)

1.3 Daftar File di motion/src/

No	Nama File	Baris	Fungsi
1	process.py	1785	State machine utama (3 role: UTAMA/SECOND/GOTO)
2	process_baru.py	1265	State machine versi lebih baru
3	proses_baru_2025.py	~2000	State machine dengan FSM lengkap + lane zones
4	proses_noncom.py	~1700	State machine tanpa komunikasi multi-robot
5	proses_noncom_nogyro.py	~1400	State machine tanpa kom dan tanpa gyro
6	process_com_beta.py	~2000	State machine dengan komunikasi beta
7	process_wilayah_shooter.py	~700	State machine untuk role shooter regional
8	process_wilayah_kickoff.py	~300	State machine untuk role kickoff regional
9	projecttes.py	~2000	Prototype/test state machine
10	mlaku.py	~1700	Duplikat proses_noncom (mlaku = jalan)
11	driver.py	587	Low-level driver (original OP3 walk)
12	driver_baru_2025.py	~2500	Driver terbaru (OP3 + Quintic walk)
13	driver_.py	~2500	Driver lama (tanpa quintic)
14	keeper_driver.py	~200	Driver khusus kiper (sederhana)
15	keeper_proses.py	0	KOSONG – belum ada logika kiper
16	odometry.py	398	Odometry original (OP3 walk)
17	odometry_baru_2025.py	444	Odometry terbaru (Quintic walk)
18	odometry_bno.py	~1500	Odometry dengan IMU BNO055
19	odometry_new.py	~700	Odometry sederhana (joint states langsung)
20	odometry_nongyro.py	~900	Odometry tanpa gyro
21	lapangan.py	136	Visualisasi lapangan Pygame
22	skema1.py	~350	Skema kickoff multi-robot 1
23	skema2.py	~300	Skema kickoff multi-robot 2
24	jalan_kotak.py	~200	Test jalan kotak
25	tesjalan.py	~200	Test jalan sederhana
26	tesskema1.py	0	KOSONG
27	test.py	2	Print hello world
28	projecttes.py	~2000	Prototype state machine

2. ARSITEKTUR MOTION SYSTEM

2.1 Diagram Aliran Data



2.2 Dual Walk Profile Architecture

Kode motion Gandamana memiliki **dua profile walking engine**:

Profile	Message Type	Topic	Digunakan di
Original OP3 Walk	op3_walking_module_msgs/WalkingParam	/robotis/walking/set_params	driver.py, odometry.py, proses_noncom_nogyro.py
Quintic Walk (Custom)	quintic_walk_msgs/WalkingParam	/robotis/quintic_walk/set_params	driver_baru_2025.py, odometry_baru_2025.py, odometry_bno.py

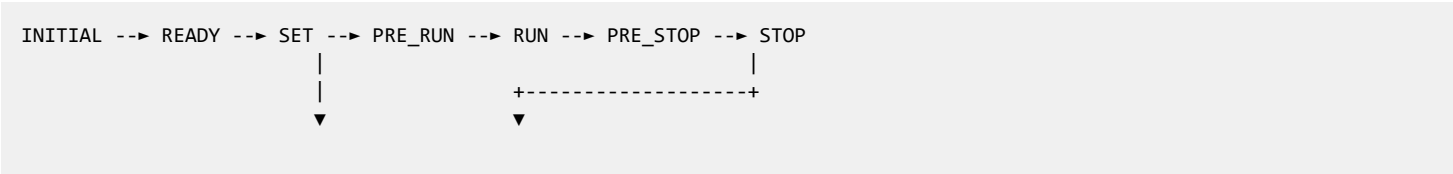
Beberapa file support **kedua profile** (driver_baru_2025.py, process_com_beta.py, projecttes.py) dengan konstanta PROFILE_QUINTIC.

2.3 Tiga Role Robot

Sistem menggunakan 3 peran yang ditentukan oleh Com_decision():

UTAMA --- Striker utama: follow_ball → positioning → kick
SECOND -- Pendukung: follow_ball → flank positioning
GOTO ---- Pemburu bola (tanpa nendang)

2.4 Game Status Lifecycle



3. ROS TOPIC MAP LENGKAP

3.1 Vision Topics

Publisher	Subscriber	Type	Keterangan
/DEW0/image_processing/deteksi_bola/ball_state	process, driver	BallState	FOUND/NOTFOUND
/DEW0/image_processing/deteksi_bola/coordinate	process, driver	BallCoordinate	(pos_x, pos_y, obj_size)
/DEW0/image_processing/deteksi_bola/goal_state	process	GoalState	Goal FOUND/NOTFOUND
/DEW0/image_processing/deteksi_bola/goal_coordinate	process	GoalCoordinate	Goal position
/DEW0/image_processing/deteksi_bola/tracking_state	driver	String	start/stop tracking
/DEW0/image_processing/deteksi_bola/positioning_state	driver	String	start/stop positioning
/DEW0/image_processing/deteksi_bola/keeper_action	driver	String	A/B/C/X keeper action

3.2 Motion Control Topics (process → driver)

Publisher (process)	Subscriber (driver)	Type	Keterangan
/DEW0/MotionControl/MotionModule	driver	String	"walk"/"action"/"head"
/DEW0/MotionControl/Command	driver	String	"start"/"stop"/"reset"
/DEW0/MotionControl/WalkingParams	driver	WalkingParam	Gait parameters
/DEW0/MotionControl/ActionNum	driver	Int32	Action page number
/DEW0/MotionControl/HeadPan	driver	Float64	Head pan angle
/DEW0/MotionControl/HeadTilt	driver	Float64	Head tilt angle
/DEW0/MotionControl/HeadScan	driver	Bool	Enable head scan
/DEW0/MotionControl/pan_tilt_error	process	Float32MultiArray	PID error feedback
/DEW0/MotionControl/yaw	odometry, lapangan	Float32	Normalized yaw heading

3.3 Hardware Topics (OpenCR → ROS)

Topic	Subscriber	Type	Keterangan
/robotis/open_cr/yaw	process	Int16	Yaw heading (0-360°)
/robotis/open_cr/imu	process, driver	Imu	IMU orientation + angular velocity
/robotis/open_cr/button	process	String	Button press event
/robotis/goal_joint_states	driver	JointState	All joint positions
/robotis/sync_write_item	semua	SyncWriteItem	LED, buzzer control

3.4 Odometry Topics

Topic	Type	Pub/Sub	Keterangan
/DEW0/Odometry/jointkaki	JointKaki	driver → odometry	Joint angles for FK

Topic	Type	Pub/Sub	Keterangan
/DEWO/Odometry/position	Pose2D	odometry → process, lapangan	(x, y, theta)
/DEWO/Odometry/goalangle	Float32	odometry → process	Heading to goal
/DEWO/Odometry/goal_position	Pose2D	process → odometry	Target position
/DEWO/Odometry/initpoint	Pose2D	process → odometry	Reset initial position
/DEWO/Odometry/task	String	process → odometry	"move"/"rotate"/"stay"
/DEWO/Odometry/cmd	String	process → odometry	"start"/"stop"

3.5 Communication Topics

Topic	Type	Pub/Sub	Keterangan
/DEWO/Communication/robotinfo	RobotInfo	receiver → process	Robot lain state
/DEWO/Communication/from_udp_to_ros	String	receiver → process	Command dari jaringan
/DEWO/Communication/from_ros_to_udp	String	process → sender	Data ke jaringan
/DEWO/Communication/Task	String	process → sender	Broadcast task
/DEWO/Communication/Bodystate	String	process → sender	Fall state

3.6 Game Controller Topics

Topic	Type	Subscriber	Keterangan
/DEWO/GameController/allstate	GameControllerState	process	GC state

3.7 Hardware Command Topics (driver → robotis)

Publisher (driver)	Topic	Type	Keterangan
pub_WalkingParams	/robotis/walking/set_params	WalkingParam	Gait parameters
pub_WalkingCommand	/robotis/walking/command	String	"start"/"stop"
pub_EnableCtrlModule	/robotis/enable_ctrl_module	String	Module selection
pub_InitPose	/robotis/base/ini_pose	String	Initial pose reset
pub_PageNum	/robotis/action/page_num	Int32	Action page execution
pub_HeadJoint	robotis/head_control/set_joint_states	JointState	Head position
pub_HeadScan	robotis/head_control/scan_command	String	"scan"/"stop"

4. process.py – STATE MACHINE UTAMA

File: motion/src/process.py | Baris: 1785 | Role: UTAMA/SECOND/GOTO

4.1 Subscribers

/robotis/open_cr/yaw	→ yaw_heading (Int16 → yaw_fixed)
/robotis/open_cr/imu	→ imu_orientation (Imu → fall detection)
/robotis/open_cr/button	→ opencr_button2 (String → mode control)
/robotis/goal_joint_states	→ present_joint_states (JointState → head pan/tilt)

```

DEWO/.../ball_state           → ball_state_status (BallState → FOUND/NOTFOUND)
DEWO/.../coordinate           → ball_coordinate (BallCoordinate → pos + size)
DEWO/.../goal_coordinate       → goal_coordinate (GoalCoordinate → goal pos)
DEWO/.../goal_state           → goal_state_status (GoalState → FOUND/NOTFOUND)
DEWO/MotionControl/pan_tilt_error → pan_tilt_error (Float32MultiArray → PID error)
DEWO/Communication/robotinfo   → RobotInfo_Callback (RobotInfo → multi-robot)
DEWO/Communication/from_udp_to_ros → from_udp_to_ros (String → remote command)
DEWO/Odometry/position         → location_Callback (Pose2D → X,Y,W global)
DEWO/Odometry/goalangle        → goalangle_Callback (Float32 → heading goal)
DEWO/GameController/allstate   → GCState_Callback (GameControllerState → status)

```

4.2 Publishers

```

/DEWO/MotionControl/yaw        # yaw_fixed2 termormalisasi
/DEWO/MotionControl/HeadPan    # Head pan angle
/DEWO/MotionControl/Command    # "start"/"stop"/"reset"
/DEWO/MotionControl/HeadScan   # Bool scan enable
/DEWO/MotionControl/HeadTilt    # Head tilt angle
/DEWO/MotionControl/ActionNum   # Action page number
/DEWO/MotionControl/MotionModule # "walk"/"action"/"head"
/DEWO/MotionControl/WalkingParams # WalkingParam
/robotis/sync_write_item        # LED & buzzer control
/DEWO/Odometry/task             # "move"/"rotate"/"stay"
/DEWO/Odometry/cmd              # "start"/"stop"
/DEWO/Odometry/goal_position    # Target position (x,y)
/DEWO/Odometry/initpoint        # Reset init position
/DEWO/Communication/Task        # Taskcom broadcast
/DEWO/Communication/Bodystate   # Fall state broadcast
/DEWO/Communication/from_ros_to_udp # Data ke jaringan
/DEWO/.../tracking_state        # start/stop tracking
/DEWO/.../positioning_state     # start/stop positioning

```

4.3 Global Variables Penting

```

Status      # Game status: READY, SET, RUN, STOP, IDLE, dll
Task         # Task saat ini: initial, follow_ball, positioning_kick, dll
robotstatus  # Role: "UTAMA", "SECOND", "GOTO"
Taskcom      # Broadcast task: "fb", "rt", "posbl", "kck", "sc", dll

yawHeading   # Raw yaw dari OpenCR (0-360°)
yaw_fixed2   # Yaw ternormalisasi (-180° sd 180°)
yawOffset    # Offset kalibrasi yaw

ball_x, ball_y, ball_size # Posisi & ukuran bola dari vision
ball_x_angle, ball_y_angle # Sudut bola (pan/tilt + error)
status_ball    # 1 = FOUND, 0 = NOTFOUND

goal_x, goal_y      # Posisi gawang dari vision
status_goal         # 1 = FOUND, 0 = NOTFOUND
goal_orientation     # Heading ke gawang (dari odometry)

X_global, Y_global, W_global # Posisi robot dari odometry
pan_present, tilt_present    # Sudut head joint (radian)
pan_error, tilt_error        # Error PID head tracking

body_status      # "fallen_forward", "fallen_backward", ""
JERSEY           # "CYAN" atau "MAGENTA"

```

4.4 Game Status Handlers

INITIAL → READY

```
if STATE == "STATE_INITIAL":
    initgate += 1
    if initgate <= 1:
        Status = 'READY'
```

READY → PRE_POSITIONING_KICKOFF

```
if STATE == 'STATE_READY':
    readygate += 1
    if readygate <= 1:
        Status = 'PRE_POSITIONING_KICKOFF'
```

SET → PRE_STOP

```
if STATE == "STATE_SET":
    setgate += 1
    if setgate <= 1:
        Status = 'SET' # langsung jadi PRE_STOP
```

PLAYING → PRE_RUN → RUN

```
if STATE == 'STATE_PLAYING':
    playgate += 1
    if playgate <= 1:
        Status = 'PRE_RUN'
```

FINISHED → PRE_STOP

```
if STATE == 'STATE_FINISHED':
    finishgate += 1
    if finishgate <= 1:
        Status = 'PRE_STOP'
```

4.5 Button Mode Flow

Running Mode (default)

Tombol	Aksi
mode	STOP→PRE_IDLE (duduk) atau IDLE→READY (berdiri)
start	STOP→PRE_RUN atau RUN→PRE_STOP (toggle)
user	Kalibrasi yaw ke 0°
start_long	Set initpoint ke "Positioning_Kick_Off"
mode_long	Masuk mode initpoint (Cyan area)
user_long	Masuk mode initpoint (Magenta area)

Initpoint Mode

Area	start	mode	user	mode_long	user_long
Cyan	CyanTengah	-	CyanBelakang	Keluar mode	CyanBelakang_seberang
Cyan (long)	-	-	-	CyanTengah_seberang	-

Area	start	mode	user	mode_long	user_long
Magenta	MagentaTengah	MagentaBelakang	-	MagentaBelakang_seberang	Keluar mode

4.6 Com_decision() – Role Assignment

```
def Com_decision():
    Alur:
    1. Jika ada robot dengan Taskcom="kck" → SECOND (hindari double-kick)
    2. Jika ball_size ≥ threshold:
        - Robot dengan ball_size terbesar → UTAMA
        - Lainnya → SECOND
    3. Jika ball_size ≥ range:
        - Bandingkan yaw_fixed2 vs posw robot lain
        - Yang lebih kecil offset yaw ke goal → UTAMA
    4. Jika self fallen → SECOND
    5. Jika teammate fallen → UTAMA
    6. Jika ada yang "rt" atau "posbl" → SECOND
    7. Jika tidak ada koneksi → UTAMA (standalone)
```

4.7 Task Flow UTAMA (Detail)

Task initial (baris 1537-1559)

- count_task += 1 tiap 1000ms
- Motion_Start(), start_tracking(True), OdomStart()
- Jika check_ball_lost(10): walk_follow_head dengan amove dari pan
- Jika tidak: jalan lurus
- setelah 3 siklus (count_task > 2) → Task = 'follow_ball'

Task follow_ball (baris 1561-1594)

- Com_decision() tiap siklus
- Taskcom = "fb"
- xmove, ymove, amove = walk_follow_head(pan_present, tilt_present)
- walking(xmove, ymove, amove, 0.63)
- If tilt_degree < -60 → Task = 'heading_to_goal_first'
- Jika ball lost 20 frame: scanning (rotasi acak kiri/kanan)

Task heading_to_goal_first (baris 1596-1619)

- Jika ball lost → kembali ke follow_ball
- Jika yaw_fixed2 dalam ±5° dari goal_orientation:
 - Goalposition(JERSEY), Task = 'positioning_kick'
- Jika tidak: walk_rotate2(goal_orientation)

Task positioning_kick (baris 1621-1674)

- Goalposition(JERSEY) untuk update goal_orientation
- Jika yaw dalam ±10°:
 - walk_ball_position_to_kick(target_pan=1.0, target_tilt=-74, heading)
 - Syarat located: delta_pan<2, delta_tilt<13, ball_size>4000
 - count_goal_found > 2 → kick_ball
- Jika yaw di luar toleransi: walk_rotate2(goal_orientation)
- Jika ball lost: scanning, Task kembali ke heading_to_goal_first

Task kick_ball (baris 1676-1712)


```

- OdomStop()
- ball_x_angle < 0 → action 230 (Right Kick)
- ball_x_angle > 0 → action 225 (Left Kick)
- Taskcom = "kck"
- Setelah kick: Motion_InitWalking(), Motion_Start()
- Task kembali ke 'initial'

```

Task get_up (baris 1713-1735)

```

- OdomStop()
- fallen_forward → action 122 (bangun depan)
- fallen_backward → action 66 (bangun belakang)
- Setelah bangun → walk_ready → initial

```

4.8 Walk Functions

walk_follow_head (baris 848)

```

pan_degree → a_move (map -89..89 ke -15..15)
abs(a_move) → x_move (map 0..15 ke 20..0)
tilt_degree → x_move_tilt (map tilt_min..0 ke x_move..-0.008)
return x_move_tilt*0.001, 0, a_move # x dalam meter

```

walk_ball_position_to_kick (baris 1199)

```

target_pan=1.0, target_tilt=-74
delta_pan = target_pan - pan_degree
delta_tilt = target_tilt - tilt_degree

x_move dari delta_tilt_pan (delta_tilt + abs(delta_pan)*0.3)
y_move dari pan_degree vs target_pan
a_move dari Heading_Gap(heading, yaw_fixed2)

located = (delta_pan > -2 dan < 2) dan delta_tilt < 13 dan ball_size > 4000

```

walk_to_position (baris 864)

```

distance = sqrt((x - X_global)² + (y - Y_global)²)
gap = Heading_Gap(goal_orientation, yaw_fixed2)

Mode 'move':
- a_move = map(gap, -180..180, -15..15)
- x_move dari abs(a_move) dan distance
- y_move = PID(gap, 0) / 30 (clamp ±15)
- Jika X dalam ±6 dari target → mode 'netral'

Mode 'netral':
- x_move = -4 (mundur sedikit)
- Rotasi halus ke target heading
- Jika yaw dalam ±6° dari target → in_position = True

```

walk_rotate2 (baris 932)

```

heading_gap = Heading_Gap(heading, yaw_fixed2)
delta_tilt = -57 - tilt_degree

x_move dari delta_tilt (maju/mundur biar dapet sudut pandang)

```

```
y_move = PID(heading_gap, 0) / 30 (clamp ±15)
a_move = map(pan_degree, -10..10, -6..6)
```

4.9 Utility Functions

Heading_Gap (baris 815)

```
Menghitung selisih sudut terpendek antara goal dan yaw.
Handle kasus 跨越 0°/180° boundary:
- Jika goal positif dan yaw negatif (atau sebaliknya):
  Hitung gap via 0° dan via 180°, pilih yang lebih pendek
- Lainnya: gap = goal_orientation - yaw_fixed2
```

calculate_pid (baris 723)

```
kp = 8.0, ki = 5.0, kd = 15.0
error = set_point - val
rateError = error - lastError
out = kp*error + kd*rateError # (ki tidak dipakai - dikomentari)
```

map_value vs map_value2

```
map_value: Dengan clamping (nilai dibatasi min/max source)
map_value2: Tanpa clamping (bisa di luar range)
```

check_ball_lost (baris 793)

```
Counter: setiap frame tanpa bola → ball_was_lost++
Jika ball_was_lost ≥ threshold → return 1 (lost)
Jika bola ketemu → reset counter
```

5. process_baru.py – STATE MACHINE VERSI BARU

File: motion/src/process_baru.py | Baris: 1265

5.1 Perbedaan dari process.py

Aspek	process.py	process_baru.py
Role system	UTAMA/SECOND/GOTO (Com_decision)	Single robot (tanpa multi-role)
Ball topic	/DEWO/.../coordinate	/v9_ball_detector_node/ball_cordinate
Komunikasi	Ada (robotinfo, from_udp_to_ros)	Tidak ada
RobotInfo	Ada (RobotInfo_Callback)	Tidak ada
Taskcom	Ada broadcast ke UDP	Tidak ada
HEADING_TO_GOAL	2 tahap (first + second)	1 tahap langsung ke positioning
Kick action	225/230 (langsung)	176-183 (sequential)
yaw handling	Filter anti-loncat 179°	Langsung, tanpa filter

5.2 Status Lifecycle

```
INITIAL → READY → STANDBY → STOP
      ↘ RUN (langsung dari READY)
        ↑ (toggle via button start)
      PRE_STOP
```

Catatan: Tidak ada PRE_POSITIONING_KICKOFF
STATE_READY → langsung RUN, Task = "Positioning"

5.3 Task Flow

```
initial (2 siklus × 1000ms)
↓
follow_ball (tilt < -50 → heading_to_goal_first)
↓
heading_to_goal_first (yaw ±15° → heading_to_goal_second)
↓
heading_to_goal_second (walk_ball_positioning_to_goal)
                        (atau yaw ±5° → positioning_kick)
↓
positioning_kick (walk_ball_position_to_kick)
↓
kick_ball (action 176-183)
↓
initial (loop)
```

5.4 Parameters Berbeda

```
walk_follow_head:
  a_move_max = 15.0
  x_move_max = 30.0 (vs 20.0 di process.py)
  tilt_min = -50 (vs -61 di process.py)

positioning_kick condition:
  (delta_pan < 4 and delta_tilt < 4) or ball_size > 3000
  (vs delta_pan > -2 dan < 2, delta_tilt < 13, ball_size > 4000)

kick_ball action:
  ball_x_angle < 0: action 180,181,182,183 (sequential)
  ball_x_angle > 0: action 176,177,178,179 (sequential)
```

5.5 Head Control yang Berbeda

```
# process_baru.py panggil Motion_HeadControl langsung
Motion_HeadControl(0.0, -0.20) # Setelah kick
# vs process.py yang set headTilt dulu
```

6. VARIAN PROCESS LAINNYA

6.1 proses_baru_2025.py

Baris: ~2000 | **Profile:** Dual (OP3 + Quintic) | **Komunikasi:** Tidak | **Gyro:** Ya

Finite State Machine lengkap dengan lane zones:

```
class FSM:
    states = ['POSITIONING', 'SearchBall', 'Tracking', 'KickBall', 'Rotate', 'GotoBall']
```

Lane zones untuk positioning:

```
# 4 lane zones berdasarkan kondisi pertandingan
lane1: Zona kiri belakang
lane2: Zona kanan belakang
lane3: Zona kiri depan
lane4: Zona kanan depan
```

Fitur:

- Mode debug dengan print
- State machine dengan transisi eksplisit
- Pengaturan walking period yang berbeda tiap state
- Dual profile support (PROFILE_QUINTIC = True/False)

6.2 proses_noncom.py

Baris: ~1700 | **Profile:** Quintic only | **Komunikasi:** Tidak | **Gyro:** Ya

Sama seperti proses_baru_2025.py tanpa komunikasi. Menggunakan quintic_walk_msgs.msg.WalkingParam secara eksklusif.

State machine:

```
POSITIONING → SearchBall → Tracking → KickBall → Rotate → GotoBall
                ↑                               |
                +-----+-----+-----+-----+
```

6.3 proses_noncom_nogyro.py

Baris: ~1400 | **Profile:** Original OP3 | **Komunikasi:** Tidak | **Gyro:** Tidak

Versi paling sederhana:

- Tidak ada komunikasi multi-robot
- Tidak ada gyro (mengandalkan yaw dari OpenCR)
- Menggunakan op3_walking_module_msgs.msg.WalkingParam
- State machine lebih sederhana

6.4 process_com_beta.py

Baris: ~2000 | **Profile:** Dual | **Komunikasi:** Ya (RobotInfo) | **Gyro:** Ya

Fitur komunikasi multi-robot:

```
rospy.Subscriber("/DEWO/Communication/robotinfo", RobotInfo, RobotInfo_Callback)
rospy.Subscriber("/DEWO/Communication/from_udp_to_ros", String, from_udp_to_ros)
```

Gate counters untuk state transitions:

```
readygate = initgate = setgate = playgate = finishgate = 0
# Mencegah multiple trigger pada state yang sama
```

Publikasi team positions:

```
# Publish posisi robot sendiri via UDP
from_ros_to_udp.publish(json_data)
```

6.5 process_wilayah_shooter.py

Baris: ~700 | **Profile:** Simplified | **Role:** Shooter

Untuk kontes regional (wilayah):

```
# Ball detection dari topic berbeda
rospy.Subscriber("/op3_ball_detector/ball_position", Pose, ball_position_Callback)

# Task list sederhana:
POSITIONING → DRIBBLING → KICKING → STOP
```

6.6 process_wilayah_kickoff.py

Baris: ~300 | **Profile:** Simplified | **Role:** Kickoff

Untuk eksekusi kickoff saja:

```
# Sangat sederhana, hanya untuk positioning + kick
Task = "POSITIONING" → "KICKING" → "STOP"
```

6.7 projecttes.py

Baris: ~2000 | **Profile:** Dual | **Komunikasi:** Ya | **Fitur:** Sequence Buffer

Prototype dengan sequencing:

```
Sequence_Buffer = []      # Buffer untuk eksekusi berurutan
Step_counter = 0          # Counter langkah dalam sequence
```

State machine lengkap dengan komunikasi multi-robot + dual walk profile.

6.8 mlaku.py

Baris: ~1700 | **Profile:** Quintic only

Duplikat dari proses_noncom.py (mlaku = "jalan" dalam bahasa Jawa).

7. driver.py – LOW-LEVEL DRIVER

File: motion/src/driver.py | **Baris:** 587

7.1 Fungsi Utama

1. Menerima perintah dari process.py via 6 topic subscriber
2. Mengontrol module robotis: walking, action, head control
3. PID head tracking untuk bola dan gawang
4. Mempublikasikan joint kaki untuk odometry

7.2 Status & Mode

```

STATUS_STARTUP = "startup"    # Inisialisasi
STATUS_ERROR   = "error"      # Error
STATUS_STOP    = "stop"       # Berhenti
STATUS_STANDBY = "standby"    # Siap
STATUS_RUN     = "run"        # Berjalan

MODE_NONE      = "none"       # Tidak ada module
MODE_WALK      = "walk"       # Walking module aktif
MODE_ACTION    = "action"     # Action module aktif
MODE_HEAD      = "head"       # Head control module aktif

```

7.3 Module Switching Logic

```

def MotionModule_Callback(data):
    if data.data == MODE_WALK:
        if status == standby/stop:
            EnableCtrlModule("walking_module")
            motionModule = MODE_WALK
    elif data.data == MODE_ACTION:
        if status == standby/stop:
            EnableCtrlModule("action_module")
            motionModule = MODE_ACTION
    elif data.data == MODE_HEAD:
        EnableCtrlModule("head_control_module")
        # head bisa diaktifkan kapan saja

```

7.4 Command Logic

```

def Command_Callback(data):
    if data.data == "start":
        if status == stop:
            WalkingCommand("start")
            taskStatus = STATUS_RUN
    elif data.data == "stop":
        WalkingCommand("stop")
        taskStatus = STATUS_STOP
    elif data.data == "reset":
        WalkingCommand("stop") # jika running
        InitPose()
        taskStatus = STATUS_STANDBY

```

7.5 PID Head Tracking

Tracking Bola (baris 192-253)

```

p_gain = 0.265
i_gain = 0.0
d_gain = 0.02

x_err = -arctan(x * arctan(w)) # w = 35.2° FOV horizontal
y_err = -arctan(y * arctan(h)) # h = 21.6° FOV vertikal

x_err_diff = (x_err - pan) / delta_time
y_err_diff = (y_err - tilt) / delta_time

x_err_target = x_err * p_gain + x_err_diff * d_gain
y_err_target = y_err * p_gain + y_err_diff * d_gain

```

Tracking Gawang (baris 255-303)

```
p_gain = 0.08      # Lebih kecil dari bola (lebih lambat)
i_gain = 0.0
d_gain = 0.028

# Error calculation sama dengan bola
```

7.6 Main Loop (baris 531-587)

```
Loop 10Hz:
1. Jika tracking_state == "start" dan ball FOUND:
    set_head_joint_offset(x_err_target, y_err_target)
2. Jika tracking_state == "start" dan ball NOTFOUND:
    head_start_scan() (setelah 20 frame lost)
3. Jika positioning_state == "start" dan goal FOUND:
    set_head_joint_offset(x_err_target, y_err_target)
4. Jika positioning_state == "start" dan goal NOTFOUND:
    head_start_scan() (setelah 5 frame lost)
5. Jika walking params berubah → publish ke robotis
6. Jika head joint berubah → publish ke robotis
7. Tiap 100ms → publish gyroYawIntegrator
```

7.7 Publishers

```
pub_JointKaki      → /DEWO/Odometry/jointkaki
pub_WalkingParams  → /robotis/walking/set_params
pub_WalkingCommand → /robotis/walking/command
pub_EnableCtrlModule → /robotis/enable_ctrl_module
pub_InitPose       → /robotis/base/ini_pose
pub_PageNum        → /robotis/action/page_num
pub_HeadJoint      → robotis/head_control/set_joint_states
pub_HeadScan       → robotis/head_control/scan_command
pub_TaskStatus     → /KRSBI/Manuvering/Status
pub_YawPos         → /KRSBI/Manuvering/Sensor/Gyro/Yaw
headoffsetpub      → /robotis/head_control/set_joint_states_offset
pantilterror      → /DEWO/MotionControl/pan_tilt_error
```

8. driver_baru_2025.py – DRIVER TERBARU

File: motion/src/driver_baru_2025.py | Baris: ~2500

8.1 Perbedaan dari driver.py

Aspek	driver.py	driver_baru_2025.py
Walk profile	Original OP3 only	Dual (OP3 + Quintic)
Walking params topic	/robotis/walking/set_params	/robotis/quintic_walk/set_params
Profile selection	Tidak ada	PROFILE_QUINTIC constant
Proses state machine	Built-in di main loop	State machine + processed
Button handling	Tidak ada	Ada (OpenCR button + mode)
Gyro yaw publish	Ada	Ada (lebih detail)

8.2 Dual Profile Support

```
PROFILE_QUINTIC = True    # Ubah ke False untuk OP3 walk

if PROFILE_QUINTIC:
    # Subscribe ke quintic_walk_msgs
    rospy.Subscriber("/robotis/quintic_walk/set_params", ...)
else:
    # Subscribe ke op3_walking_module_msgs
    rospy.Subscriber("/robotis/walking/set_params", ...)
```

8.3 Walking Parameters (Quintic Profile)

```
WalkingParam_msg = WalkingParam()
# Parameter spesifik untuk quintic walk engine:
# trunkXOffset, trunkYOffset, trunkPitch, setpointPitch
# YMove, XMove, period_time, doubleSupportRatio
```

8.4 Fitur Tambahan

- **Control loop rate:** 10Hz
- **Publish multiple status topics**
- **Subscriber untuk processed command** (selain dari motion module)
- **Joint state monitoring** lebih detail
- **Button handling** untuk mode running dan initpoint

8.5 driver_.py

File: motion/src/driver_.py | **Baris:** ~2500

Hampir identik dengan driver_baru_2025.py tetapi:

- **Tidak memiliki konstanta PROFILE_QUINTIC**
- Hanya support original OP3 walk profile
- Kemungkinan versi sebelum quintic walk diimplementasikan

9. VARIAN DRIVER LAINNYA

9.1 keeper_driver.py

File: motion/src/keeper_driver.py | **Baris:** ~200

Driver yang disederhanakan untuk **keeper (kiper)**:

Perbedaan dari driver.py:

```
# Tidak punya:
- WalkingParam subscriber (tidak butuh gait tuning)
- Odometry goal_position subscriber
- Quintic walk profile support
- ActionNum subscriber (maybe - tergantung implementasi)

# Punya:
```


- gyroYawIntegrator publish
- Ball tracking PID (sederhana)

Penting: Driver kiper sudah ada, tapi **keeper_proses.py masih kosong** – belum ada logika strategi kiper.

10. ODOMETRY SYSTEM

10.1 Prinsip Dead-Reckoning

Odometry Gandamana menggunakan **kinematika kaki** (forward kinematics) untuk menghitung perpindahan robot. Tidak menggunakan visual odometry atau encoder roda.

Setiap langkah:

1. Baca sudut joint kaki → ForwardOdom() → displacement (X, Y, Angle)
2. Akumulasi ke posisi global → position()
3. Kirim ke process.py via Pose2D

10.2 odometry_baru_2025.py (Terbaru, Direkomendasikan)

File: motion/src/odometry_baru_2025.py | **Baris:** 444

WalkingParam dari Quintic Walk

```
rospy.Subscriber("/robotis/quintic_walk/set_params", WalkingParam, WalkingParam_Callback)
# Parameter:
x_offset = data.trunkXOffset * 100    # m → cm
y_offset = data.trunkYOffset * 100
p_offset = data.trunkPitch
hip_offset = data.setpointPitch
swim = data.YMove                      # Untuk deteksi arah side step
```

Forward Kinematics (ForwardOdom)

```
P = 12    # paha (cm)
B = 12    # betis (cm)
K = 3.05  # telapak kaki (cm)

def ForwardOdom(PG, LT, GL):
    # PG = pinggul, LT = lutut, GL = kaki
    Hr = P*cos(PG) + B*cos(PG+LT) + K*cos(PG+LT-GL)  # tinggi kaki kanan
    Hl = P*cos(-PG) + B*cos(-PG-LT) + K*cos(-PG-LT+GL) # tinggi kaki kiri
    Xr = P*sin(PG) + B*sin(PG+LT) + K*sin(PG+LT-GL) - x_offset
    Xl = P*sin(-PG) + B*sin(-PG-LT) + K*sin(-PG-LT+GL) - x_offset
    return Hr, Hl, Xr, Xl
```

Side Kinematics (SideOdom)

```
def SideOdom(PG, GL, PB):
    base = PB * sin(-PG)
    step = K * sin(-PG + GL)
    Yr = base + step - y_offset/2
    Yl = base + step + y_offset/2
    return Yr, Yl
```

Angle Kinematics (AngleOdom)

```
def AngleOdom(Rpg, Lpg):
    Angle = abs(Rpg) + abs(Lpg)
    return Angle
```

XMove & YMove (Akumulasi Displacement)

```
def XMove(Swap, Move, Angle):
    noise = 1.3
    swapX = -Swap
    X = moveX + swapX + noise
    moveX = Move
    return X, Angle

def YMove(supportY, swingY, Angle):
    Y = swingY - supportY
    return Y, Angle
```

Position Accumulation

```
def position(X, Y, W):
    W_global = deg2rad(W_global)
    # XMOVE
    X_global += X * cos(W_global)
    Y_global -= X * sin(W_global)
    # YMOVE
    X_global += Y * cos(pi/2 + W_global)
    Y_global -= Y * sin(pi/2 + W_global)
    position_Pub(X_global, Y_global, rad2deg(W_global))
```

Phase-Locked Trigger

```
# Hanya hitung saat fase ganti (mencegah double-counting)
if fase in ["1", "2", "3"]:
    trigger_odometri = True

if trigger_odometri:
    if fase == "2": leg = 0 # kaki kanan
    else: leg = 1          # kaki kiri
    # Hitung displacement sesuai task
    position(X_curr, Y_curr, W_curr)
```

Goal Angle Calculation

```
def goalposition_Callback(data):
    delta_x = data.x - X_global
    delta_y = data.y - Y_global
    w_goal = arctan2(-delta_y, delta_x)
    # -delta_y karena koordinat Y terbalik dengan pygame
    GoalAnglePub.publish(rad2deg(w_goal))
```

Task Modes

```
"move":      ForwardOdom + XMove (maju/mundur)
"rotate":    SideOdom + YMove (geser sampling + rotasi)
```

```
"stay":      AngleOdom aja (diam, tracking rotasi aja)
"positioning": Semua komponen (move + rotate + side)
```

10.3 odometry.py (Original)

File: motion/src/odometry.py | Baris: 398

Perbedaan dari odometry_baru_2025.py

Aspek	odometry.py	odometry_baru_2025.py
Walking param from	/robotis/walking/set_params	/robotis/quintic_walk/set_params
WalkingParam msg	op3_walking_module_msgs	quintic_walk_msgs
Parameter naming	init_x_offset, dsp_ratio	trunkXOffset, doubleSupportRatio
Phase topic	/DEW0/phase	/DEW0/Walking/Phase
Phase trigger	Manual (count)	Otomatis (trigger_odometri)
YMove	Sederhana	Dengan swim detection

WalkingParam OP3 vs Quintic

OP3 (odometry.py) :

```
x_offset = data.init_x_offset * 100
y_offset = data.init_y_offset * 100
p_offset = data.init_pitch_offset
hip_offset = data.hip_pitch_offset
periode = data.period_time
DSP_ratio = data.dsp_ratio
```

Quintic (odometry_baru_2025.py) :

```
x_offset = data.trunkXOffset * 100
y_offset = data.trunkYOffset * 100
p_offset = data.trunkPitch
hip_offset = data.setpointPitch
swim = data.YMove # untuk deteksi arah
DSP_ratio = data.doubleSupportRatio
```

YMove yang Berbeda

odometry.py:

```
YMove(-L_side, R_side, -Angle) # leg kiri
YMove(-R_side, L_side, Angle)  # leg kanan
```

odometry_baru_2025.py:

```
# Tergantung arah swim (side step)
if swim_cek == "kiri":
    YMove(R_side, L_side, -Angle)
elif swim_cek == "kanan":
    YMove(L_side, R_side, Angle)
```

10.4 odometry_bno.py

File: motion/src/odometry_bno.py | **Baris:** ~1500**Fitur**

- Menggunakan **IMU BNO055** untuk heading feedback
- Subscribe ke /DEWO/Odometry/jointkaki (JointKaki msg)
- Menggunakan quintic_walk_msgs.msg.WalkingParam
- Publikasi Pose2D odometry dan Float32 yaw

Kelebihan

- Heading dari IMU lebih akurat daripada dead-reckoning murni
- BNO055 memiliki sensor fusion internal (accelerometer + gyro + magnetometer)

Kekurangan

- Lebih kompleks (1500 baris)
- Bergantung pada IMU yang bisa kena drift magnetik

10.5 odometry_new.py**File:** motion/src/odometry_new.py | **Baris:** ~700**Fitur**

- Membaca /robotis/goal_joint_states **langsung** (tanpa JointKaki msg)
- Tanpa ketergantungan IMU
- Lebih sederhana dari odometry_bno.py

Kelebihan

- Sederhana, mudah di-debug
- Tidak butuh IMU

Kekurangan

- Akurasi lebih rendah (tanpa feedback IMU)

10.6 odometry_nongyro.py**File:** motion/src/odometry_nongyro.py | **Baris:** ~900**Fitur**

- FK sama dengan odometry_bno.py
- Heading dari **yaw eksternal** (OpenCR / topic lain)
- Tidak menggunakan gyro/IMU internal
- Menggunakan quintic_walk_msgs.msg.WalkingParam

11. lapangan.py – FIELD VISUALIZATION**File:** motion/src/lapangan.py | **Baris:** 136

11.1 Spesifikasi

Aspek	Nilai
Library	Pygame
Resolusi	1040 × 660 pixel
FPS Target	100
Judul	"KRSBI-H: 'GANDAMANA' UNIVERSITAS NEGERI SURABAYA"
Skala (P)	1 (1 unit = 1 pixel)

11.2 Dimensi Lapangan

Elemen	Internal	Visual (P=1)
Panjang field	900	900 px
Lebar field	600	600 px
Goal Area width	100	100 px
Goal Area height	500	500 px
Center circle radius	75	75 px
Penalty radius	210	210 px
Goal width	260	260 px
Goal depth	60	60 px
Line width	6	6 px
Offset X (nolX)	-	69 px
Offset Y (nolY)	-	30 px

11.3 Warna & Area

```

CYAN area    = Rect(0, 0, 520, 660)    # Setengah kiri
MAGENTA area = Rect(520, 0, 520, 660)  # Setengah kanan
Background   = dark red
Field        = green (0, 200, 0)
Lines        = white
Cyan accent  = blue (atas field)
Magenta accent = red (atas field)
Juri line    = black (bawah field)

```

11.4 Elemen yang Digambar

1. Background dark red
2. Area CYAN (biru) + MAGENTA (merah)
3. Background gawang + lapangan (hijau)
4. Garis tepi atas: biru (CYAN) + merah (MAGENTA)
5. Garis tepi bawah: hitam (juri)
6. Garis samping kiri & kanan (putih)
7. Garis tengah (putih)
8. Goal Area kiri & kanan (putih)
9. Lingkaran tengah (putih)
10. Gawang kiri & kanan dengan kedalaman 60px (putih)
11. Titik tengah (center spot)
12. Titik penalty kiri & kanan

13. Posisi robot (gambar PNG di-rotate sesuai yaw)

11.5 Konversi Koordinat

```
def position_Callback(data):
    Xglobal = (data.x / P) + nolX    # internal → pixel
    Yglobal = (data.y / P) + nolY
    Wglobal = data.theta

def position2(x, y, t):
    robot = pg.image.load("klipartz.com.png")
    pose = pg.transform.rotate(robot, yaw)
    layar.blit(pose, [x + xnol, y + ynol])
```

11.6 Subscribers

```
/DEWO/Odometry/position    → position_Callback (Pose2D)
/DEWO/MotionControl/yaw    → heading_Callback (Float32)
```

11.7 Cara Menjalankan

```
# Dengan ROS
roslaunch kri2027 lapangan.py

# Atau langsung (pastikan roscore running)
python motion/src/lapangan.py
```

12. KEEPER SYSTEM

12.1 keeper_driver.py

File: motion/src/keeper_driver.py | **Baris:** ~200

Driver sederhana untuk kiper. Tidak memiliki:

- WalkingParam subscriber (kiper tidak jalan kompleks)
- Odometry goal_position subscriber
- Quintic walk profile

Topik yang dipublish:

```
pub_JointKaki    → /DEWO/Odometry/jointkaki
pub_TaskStatus   → /KRSBI/Manuvering/Status
pub_YawPos       → /KRSBI/Manuvering/Sensor/Gyro/Yaw
```

12.2 keeper_proses.py

File: motion/src/keeper_proses.py | **Baris:** 0 (KOSONG!)

Belum ada implementasi strategi kiper!

Yang seharusnya ada (rekomendasi):

- Deteksi arah bola
- Pergerakan menyamping (side step) mengikuti bola
- Dive (loncat) ke kiri/kanan saat bola dekat
- Positioning di tengah gawang
- Koordinasi dengan role UTAMA (lempar bola setelah ditangkap)

13. MOVEMENT SCHEMES (SKEMA)

13.1 skema1.py – Kickoff Scheme 1

File: motion/src/skema1.py | **Baris:** ~350

Skema kickoff multi-robot versi 1:

```
WAIT_ROBOT2_TIME = 3000 # 3 detik

# Robot 1 = Kicker
# Robot 2 = Receiver (passing target)

Alur:
1. Robot 1 positioning di bola
2. Robot 2 positioning di posisi receiver
3. Robot 1 kick ke arah Robot 2
4. Robot 2 menerima bola
```

Publishes:

```
/KRSSI/BallPossession # Indikator siapa yang pegang bola
```

13.2 skema2.py – Kickoff Scheme 2

File: motion/src/skema2.py | **Baris:** ~300

Skema kickoff multi-robot versi 2 (alternatif):

```
WAIT_ROBOT2_TIME = 1000 # 1 detik (lebih cepat dari skema 1)

# Robot 1 = Kicker
# Robot 2 = Receiver

Alur: Sama seperti skema1 tapi timing berbeda
```

14. FILE TESTING

14.1 jalan_kotak.py

File: motion/src/jalan_kotak.py | **Baris:** ~200

Test: Robot berjalan membentuk kotak.

```

Alur:
1. Jalan maju X detik
2. Belok 90°
3. Ulangi 4 kali (kotak)

```

14.2 tesjalan.py

File: motion/src/tesjalan.py | **Baris:** ~200

Test: Jalan lurus dengan koreksi heading.

```

# Menggunakan sync_write_item untuk kontrol langsung
rospy.Publisher("/robotis/sync_write_item", SyncWriteItem, queue_size=1)

# Jalan maju dengan koreksi yaw
walking(x, 0, heading_correction, period)

```

14.3 tesskema1.py

File: motion/src/tesskema1.py | **Baris:** 0 (KOSONG!)

Tidak ada implementasi.

14.4 test.py

File: motion/src/test.py | **Baris:** 2

```
print("helloworld")
```

Test paling sederhana – hanya print.

15. COMPLETE FUNCTION REFERENCE

15.1 Motion Control Functions

MotionModule Switching

```

MotionMode(mode)           # Set module: "walk"/"action"/"head"
Motion_InitWalking()       # Switch ke walking module + delay 0.5s
Motion_InitAction()        # Switch ke action module + delay 0.5s
Motion_InitHead()          # Switch ke head module + delay 0.3s

```

Motion Command

```

MotionCommand(cmd)         # "start"/"stop"/"reset"
Motion_InitPose()          # Reset to initial pose (4s delay)
Motion_Start()             # Start walking engine
Motion_Stop()              # Stop walking engine + delay 0.5s

```

Walking


```
walking(x, y, o, t)      # Set walking params:
                        #   x: forward amplitude ( $\pm 0.020$ )
                        #   y: side amplitude ( $\pm 0.015$ )
                        #   o: rotation amplitude ( $\pm 15$ )
                        #   t: period time (0.63 normal / 0.45 after kick)
```

Action

```
Motion_ActionNum(num)    # Execute action page (delay 3-4s)
Motion_Kick(num)         # Stop → InitAction → ActionNum → InitWalking
```

Head Control

```
Motion_HeadControl(pan, tilt) # Set head position (radians)
Motion_HeadScan(enable)      # Enable/disable auto head scan
```

LED & Buzzer

```
LED_RGB(r, g, b)         # Set RGB LED (0-31 per channel)
LED_Status(r, g, b)      # Set status LED (0-1 per channel)
Buzzer(freq)             # Set buzzer frequency
BuzzerTone(count, period) # Buzzer pattern (count × period ms)
```

15.2 Odometry Control

```
OdomStart()              # Start odometry computation
OdomStop()               # Stop odometry computation
Odom_Command(command)    # Send command: "start"/"stop"
```

Target Setting

```
Pointposition(x, y)      # Set target position (Pose2D)
Goalposition(jersey)     # Set goal position based on jersey:
                        #   CYAN → (900, 300)
                        #   MAGENTA → (0, 300)
```

Navigation

```
walk_to_position(x, y, o) # Navigate to (x,y) then rotate to heading o
  returns (x_move, y_move, a_move, in_position)

walk_follow_head(pan, tilt) # Walk following ball head tracking
  returns (x_move, y_move, a_move)

walk_ball_position_to_kick(target_pan, target_tilt, heading)
  returns (x_move, y_move, a_move, is_located)

walk_rotate2(heading)     # Rotate to heading while adjusting distance
  returns (x_move, y_move, a_move)

walk_rotate(heading, y_move_change) # Simple rotate
  returns (x_move, y_move, a_move, is_done)

walk_rotate_goal(heading) # Rotate to goal heading
  returns (x_move, y_move, a_move)
```

```
walk_dribble(heading)      # Dribble walk (slow forward + side adjust)
    returns (x_move, y_move, a_move)

walk_in_ball_area(heading) # Walk within ball area
    returns (x_move, y_move, a_move)

walk_ball_positioning_to_goal(heading)
    returns (x_move, y_move, a_move, ball_located)
```

15.3 Helper Functions

Math Utilities

```
normalize_deg(angle)      # Normalize to 0-360°
Heading_Gap(goal, yaw)   # Shortest angular distance (-180 to 180)
map_value(val, smin, smax, tmin, tmax) # Map with clamping
map_value2(val, smin, smax, tmin, tmax) # Map without clamping
calculate_pid(val, set_point) # PID controller
limit_value(val, min, max)   # Clamp value
filter_value(val, coeff)     # Low-pass filter
scale_value(val, src, trg)   # Linear scale
min_value(a, b)              # Minimum
max_value(a, b)              # Maximum
```

Detection Helpers

```
check_ball_lost(threshold) # Returns 1 if ball lost for N frames
check_goal_lost(threshold) # Returns 1 if goal lost for N frames
```

Vision Control

```
start_tracking(state)      # True=start, False=stop ball tracking
start_positioning(state)   # True=start, False=stop goal positioning
```

Communication

```
udp_sendmsg(msg)           # Send message via UDP
```

Timing

```
Ticks()                   # Current time in milliseconds
```

15.4 Process Helper Functions

```
print_once(statement)     # Print only if statement changed (reduce spam)
```

16. PARAMETER TUNING GUIDE

16.1 Walking Parameters (driver.py baris 489-516)

Parameter	Nilai	Efek Menaikkan
init_x_offset	-0.003	Robot lebih maju (berat ke depan)
init_y_offset	0.038	Geser stance ke kanan
init_z_offset	0.025	Badan lebih tinggi
hip_pitch_offset	15°	Condong ke depan (stabil)
dsp_ratio	0.29	Lebih stabil (2 kaki lebih lama)
step_fb_ratio	0.28	Langkah lebih panjang
z_move_amplitude	0.09	Angkat kaki lebih tinggi
balance_enable	True	Balance aktif
balance_hip_roll_gain	0.35	Koreksi hip roll lebih kuat
balance_knee_gain	0.30	Koreksi knee lebih kuat
balance_ankle_roll_gain	0.70	Koreksi ankle roll lebih kuat
balance_ankle_pitch_gain	1.0	Koreksi ankle pitch lebih kuat
period_time	0.63	Lebih lambat (stabil)

16.2 Dinamis Walking (dari process.py)

Parameter yang dikirim tiap siklus:

```
walking(x_move, y_move, angle_move, period_time)
```

- x_move: ± 0.020 maks (maju) / -0.008 (mundur)
- y_move: ± 0.015 maks (side step)
- angle_move: ± 15 maks (rotasi)
- period_time: 0.63 (normal), 0.45 (setelah kick)

16.3 PID Parameters

Ball Tracking (driver.py)

```
p_gain = 0.265 # Proporsional
i_gain = 0.0   # Integral (tidak dipakai)
d_gain = 0.02  # Derivatif
```

Goal Tracking (driver.py)

```
p_gain = 0.08 # Lebih kecil = tracking lebih lambat
i_gain = 0.0
d_gain = 0.028
```

Walk PID (process.py)

```
kp = 8.0      # Proporsional untuk rotasi
ki = 5.0      # Integral (dikomentari)
kd = 15.0     # Derivatif
```

16.4 Thresholds

Ball Detection (process.py)

```
ballsize_treshold = 380    # Threshold jarak bola (role decision)
ballsize_range = 1000     # Range untuk advantage filter
```

Ball Lost

```
check_ball_lost(20)       # 20 frame = ~2 detik (10Hz loop)
```

Position Location

```
delta_pan < 2.0           # Toleransi pan untuk kick positioning
delta_tilt < 13            # Toleransi tilt untuk kick
ball_size > 4000          # Ukuran bola minimal untuk kick
```

Fall Detection (process.py)

```
pitch > 60° → fallen_forward
pitch < -60° → fallen_backward
```

16.5 Action Numbers

Aksi	Number	File
Kick Kanan	230	process.py (baru)
Kick Kiri	225	process.py (baru)
Kick Kanan (old)	180, 181, 182, 183	process_baru.py
Kick Kiri (old)	176, 177, 178, 179	process_baru.py
Get Up Forward	122	Semua file
Get Up Backward	66	Semua file

16.6 Koordinat Initial Position

Position	X	Y	Theta
CyanBelakang	200	600	90°
CyanTengah	350	600	90°
CyanBelakang_seberang	200	0	-90°
CyanTengah_seberang	350	0	-90°
MagentaBelakang	700	600	90°
MagentaTengah	550	600	90°
MagentaBelakang_seberang	700	0	-90°
MagentaTengah_seberang	550	0	-90°
Positioning_Kick_Off (Cyan)	100	300	0°
Positioning_Kick_Off (Magenta)	800	300	180°

17. DEBUGGING & TROUBLESHOOTING

17.1 Cheatsheet Cepat

Masalah	Cek di	Baris
Robot tidak jalan	driver.py Command_Callback	75-80
Robot jalan tapi oleng	driver.py Walking params	489-516
Robot tidak lihat bola	process.py check_ball_lost	793-801
Posisi odometry ngaco	odometry_baru_2025.py position()	150-166
Head tracking tidak stabil	driver.py PID gains	224-227
Robot tidak tendang	process.py positioning_kick condition	1644-1654
Robot jatuh terus	process.py fall detection threshold	370-377
Komunikasi multi-robot	process.py RobotInfo_Callback	522-569
Gyro yaw error	process.py yaw_heading filter	263-295

17.2 Cara Debugging

1. Cek ROS Topic

```
# Lihat output sensor
rostopic echo /robotis/open_cr/yaw
rostopic echo /robotis/open_cr/imu

# Lihat keputusan state machine
rostopic echo /DEW0/MotionControl/WalkingParams
rostopic echo /DEW0/Odometry/position

# Lihat vision
rostopic echo /DEW0/image_processing/deteksi_bola/ball_state
rostopic echo /DEW0/image_processing/deteksi_bola/coordinate
```

2. Cek ROS Graph

```
rqt_graph
# Atau
roslaunch rqt_graph rqt_graph
```

3. Logging

```
# Di process.py, gunakan:
rospy.loginfo("Pesan")      # Info
rospy.logwarn("Pesan")      # Warning
rospy.logerr("Pesan")       # Error

# Atau print_once() untuk mengurangi spam
```

4. Simulasi Tanpa Robot

```
# 1. roscore
roscore

# 2. lapangan (visual)
```

```
roslaunch kri2027 lapangan.py
```

```
# 3. Publish manual
rostopic pub /DEWO/Odometry/position geometry_msgs/Pose2D "x: 450 y: 300 theta: 90"
```

5. Kalibrasi Yaw

```
# Button user → set yawOffset = -yawHeading
# Sehingga yaw_fixed = 0 saat menghadap depan
```

17.3 Common Issues & Solutions

Robot Jalan Miring

```
Penyebab: init_y_offset atau balance_hip_roll_gain salah
Solusi: Tuning init_y_offset (positif = koreksi ke kiri)
        Naikkan balance_hip_roll_gain untuk koreksi otomatis
```

Robot Sering Jatuh ke Depan

```
Penyebab: hip_pitch_offset terlalu kecil
Solusi: Naikkan hip_pitch_offset (15° → 18°)
        Turunkan x_move_amplitude
```

Robot Tidak Merespon Vision

```
Penyebab: tracking_state tidak "start"
        ball_state callback error
Solusi: Cek start_tracking(True) dipanggil
        Cek /DEWO/.../ball_state topic
```

Odometry Drift Cepat

```
Penyebab: Noise pada FK, walking di permukaan tidak rata
Solusi: Gunakan odometry_bno.py (dengan IMU feedback)
        Tambah periode update
```

Double Kick (2 Robot Nendang Bersamaan)

```
Penyebab: Com_decision() tidak filter Taskcom="kck"
Solusi: Pastikan baris 453-458 aktif:
        if Taskcom == "kck": robotstatus = "SECOND"
```

18. LAMPIRAN: KODE AKSES CEPAT

18.1 Navigation Cheatsheet

```
# Baris penting di process.py:
1322 # main loop dimulai
```

```

1336 # Status READY handler
1351 # PRE_POSITIONING_KICKOFF
1357 # SET handler
1360 # PRE_RUN handler
1373 # PRE_STOP handler
1397 # RUN state dimulai (pemisahan role)
1507 # Task flow UTAMA dimulai
1537 # Task 'initial'
1561 # Task 'follow_ball'
1596 # Task 'heading_to_goal_first'
1621 # Task 'positioning_kick'
1676 # Task 'kick_ball'
1713 # Task 'get_up'
445 # Com_decision()
848 # walk_follow_head()
1199 # walk_ball_position_to_kick()

```

```

# Baris penting di driver.py:
531 # main loop dimulai
540-584 # Main loop logic
489-516 # Walking parameters (AKTIF)
192-253 # PID ball tracking
255-303 # PID goal tracking

```

```

# Baris penting di odometry_baru_2025.py:
278 # main loop dimulai
87-96 # ForwardOdom (kinematika)
105-110 # SideOdom (kinematika sampling)
112-115 # AngleOdom (kinematika rotasi)
125-132 # XMove (akumulasi X)
144-147 # YMove (akumulasi Y)
150-166 # Position (akumulasi global)

```

18.2 Startup Commands (Full System)

```

# Terminal 1: Manager
roslaunch op3_manager op3_manager.launch

# Terminal 2: Vision
roslaunch vision2027 vision_2027.py

# Terminal 3: Communication Sender
roslaunch communication2027 sendernasional.py

# Terminal 4: Communication Receiver
roslaunch communication2027 com_receive_nasional.py

# Terminal 5: State Machine
roslaunch kri2027 process.py

# Terminal 6: Low-Level Driver
roslaunch kri2027 driver.py

# Terminal 7: Field Display
roslaunch kri2027 lapangan.py

# Terminal 8: Game Controller
roslaunch protocolgc receiver8.py

# Terminal 9: Odometry
roslaunch kri2026 odometry_baru_20.py

# Trigger Start
rostopic pub /robotis/open_cr/button std_msgs/String "data: 'start'"

```

18.3 Quick Test Commands

```
# Test vision
rostopic echo /DEWO/image_processing/deteksi_bola/ball_state
rostopic echo /DEWO/image_processing/deteksi_bola/coordinate

# Test motion
rostopic pub /DEWO/MotionControl/Command std_msgs/String "data: 'start'"
rostopic pub /DEWO/MotionControl/WalkingParams ...

# Test odometry
rostopic echo /DEWO/Odometry/position
rostopic pub /DEWO/Odometry/goal_position geometry_msgs/Pose2D "x: 900 y: 300 theta: 0"

# Test button
rostopic pub /robotis/open_cr/button std_msgs/String "data: 'start'"
rostopic pub /robotis/open_cr/button std_msgs/String "data: 'mode'"
rostopic pub /robotis/open_cr/button std_msgs/String "data: 'user'"

# Test communication
rostopic echo /DEWO/Communication/robotinfo
rostopic pub /DEWO/Communication/from_udp_to_ros std_msgs/String "data: 'test'"
```

18.4 Istilah Penting (Glossary)

Istilah	Arti
FK / Forward Kinematics	Perhitungan posisi kaki dari sudut joint
Dead-Reckoning	Estimasi posisi tanpa referensi eksternal
Phase	Fase walking (0/1/2/3 = foot touch/swap/lift)
DSP	Double Support Phase (2 kaki di tanah)
SSP	Single Support Phase (1 kaki di tanah)
Action Page	Gerakan pre-programmed (kick, getup)
Gait	Pola jalan
Yaw	Sudut orientasi robot (kompas)
Pitch	Sudut kemiringan depan-belakang
Pan	Gerakan head horizontal (kiri-kanan)
Tilt	Gerakan head vertikal (atas-bawah)
Taskcom	Task code yang di-share antar robot via UDP
Initpoint	Posisi awal robot di lapangan
Gate Counter	Counter untuk mencegah multiple trigger state

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